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Uphill Walking Unveils Hidden Instability and Biomechanical Adaptations in Anterior Cruciate Ligament-Deficient Copers

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Abstract

Objective: Using computerised motion analysis, the study investigated possible joint instability and biomechanical adaptations during uphill walking in participants with anterior cruciate ligament deficiency (ACLD) classified as copers based on level walking performance.

Methods: Thirteen participants with ACLD and thirteen healthy adults each walked and ascended a 3-meter slope with inclines of 3°, 6°, and 9°. Trunk angular motions, lower-limb joint angles, forces and moments, and tibial inclination angles were calculated. Two-way analyses of variance were used to study between-subject (group) and within-subject (slope angle) effects on the variables, using a significant level of $\alpha = 0.05$ for all tests.

Results: The ACLD group displayed a compensatory trunk anterior tilt strategy over the stance phase of uphill walking to reduce knee extensor moments and posterior knee forces to accommodate knee instability during uphill walking. Compared to Control, the ACLD group showed increased anterior forces during single-limb support (SLS) and decreased posterior forces at the affected knee during the double-limb support (DLS) of uphill walking. They also demonstrated an increased anterior tibial angle of the affected limb with a decreased extensor moment at the affected knee around the contralateral heel-strike. The trunk anterior tilt was also accompanied by increased extensor moments at both hips and ankle dorsiflexion during terminal DLS.

Conclusions: This study demonstrates that uphill walking uncovers hidden functional instability at the ACL-deficient knee that remains undetected during level walking. This suggests that assessments based solely on level walking may overestimate the functional stability of the ACLD knee during more demanding uphill walking. Incorporating uphill walking into clinical evaluations may help identify compromised movement patterns that could necessitate surgical intervention in ACLD.

Introduction

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The anterior cruciate ligament (ACL) is crucial in guiding knee motion and resisting the tibial anterior translation and internal rotation [1]. Injuries to the ACL, often occurring in football, soccer, and basketball players [2], alter lower extremity kinematics and kinetics during activities [2-5]. Individuals with ACL deficiency (ACLD) may suffer from residual degraded muscle strength and dynamic balance despite returning to close-to-normal gait five months following ACL rupture [6, 7]. Early surgical intervention is generally superior to delayed ACL reconstruction (ACLR) for individuals requiring ACLR [8]. Therefore, accurate decision-making and optimal timing for ACLR are critical.

Traditionally, surgical decisions have relied on subjective evaluations of functional knee instability, such as recurrent giving-way episodes [9]. However, an objective approach based on specific gait deviations offers a more precise classification [10]. According to this method, individuals with ACLD who exhibit decreased knee extensor moments or altered joint loading patterns are classified as "noncopers," whereas those who maintain functional stability and resume pre-injury activity levels without compensatory postural or joint loading adjustments are termed "copers" [11]. Knee functional instability during gait differs from passive instability, which occurs when the knee's passive structures, such as ligaments, cannot resist excessive displacement. Functional stability, in contrast, refers to the dynamic control of the knee, relying on both passive structures and surrounding muscles to maintain joint stability during movement. When passive stability is compromised, as in ACLD, the knee becomes mechanically unstable, requiring compensatory muscular actions, such as increased hamstring activity, to generate posterior forces that counteract excessive anterior tibial translation caused by quadriceps activation or external loads [12].

Functional instability is typically identified through a combination of clinical tests, self-reported outcomes, and functional assessments. In ACLD, adequate hamstring strength and activation are essential for anterior-posterior knee stability. Rehabilitative training to achieve an appropriate

hamstrings-to-quadriceps strength ratio [13] enables individuals to maintain knee stability during level walking without compensatory adjustments in other joints. Such individuals are classified as copers without needing to limit quadriceps activation. In contrast, reduced knee extensor moments during level walking in noncopers reflect compensatory quadriceps avoidance, aimed at preventing anterior tibial translation that cannot be counteracted due to insufficient hamstring strength or coordination. Nonetheless, some copers may still experience difficulty with daily or sport-specific activities and eventually require ACL reconstruction [14-17]. These findings indicate that classifications based solely on subjective assessments or level walking deviations may not fully capture dynamic functional instability. Evaluating kinematic and kinetic adaptations during more challenging tasks, such as uphill walking, may provide a more sensitive assessment of true functional instability, reveal compensatory strategies, and underscore the critical role of targeted hamstrings/quadriceps training in ACLD rehabilitation.

Previous studies indicate that ACLD copers may have difficulties coping with more challenging tasks, such as stair climbing and uphill walking, which require greater moments and shear forces at the knee [18] and maintain unaltered knee contributions to the total support moment for whole-body support by a compensatory mechanism [19]. Sloped surfaces are commonplace in our living environment, making incline and decline navigation a frequent part of routine locomotion [18, 20]. During uphill walking, the stance-limb joints must generate greater muscular moments to prevent collapse and maintain dynamic balance under increased gravitational shear forces while moving the body forward and upward [20]. This activity challenges ACLD knees due to increased ankle dorsiflexion, tibial inclination, and anterior ground reaction forces (GRFs) [21]. Previous studies have reported that adults with ACLD displayed altered lower-limb joint kinematics and kinetics, such as reduced knee extensor moment, during level walking and stair climbing [2, 22, 23]. However, similar studies on uphill walking on various slopes have not been documented. On the other hand, trunk motions have been

shown to help modulate the loads at the lower limb joints during level and slope walking in healthy young adults [24, 25], but little attention has been given to the control of the trunk and its interaction with the lower limb joints in ACLD during uphill walking.

The current study aimed to investigate trunk motions, lower-limb joint kinematics, and kinetics during uphill walking in individuals with ACLD classified as copers based on level walking performance. It was hypothesised that these individuals would exhibit altered lower-limb joint kinematics and kinetics, including reduced knee extensor moments, altered hip or ankle movements, and increased trunk anterior tilt in individuals with ACLD during uphill walking, with effects influenced by slope angles. The current study seeks to evaluate the adequacy of level walking assessments for surgical decision-making in ACLD, highlighting the necessity of incorporating more challenging activities into clinical evaluations.

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Materials and Methods

Participants

Thirteen participants with unilateral anterior cruciate ligament deficiency (ACLD group; 7 men and 6 women, age: 41.8±12.6 years, height: 167.0±8.2 cm, mass: 71.0±12.5 kg) and thirteen healthy adults (Control group; 9 men and 4 women, age: 37.2±15.1 years, height: 169.8±8.8 cm, mass: 69.0±12.3 kg) were recruited from the university hospital. Both groups were matched for age, gender, height, and body mass. All participants provided informed written consent following the approval of the Institutional Research Board (IRB Permit number: N202004053). The diagnosis of unilateral ACLD was made by a senior orthopaedic surgeon (J.Y.W.) according to the following: (1) a positive Lachman's test and (2) a complete tear of the unilateral ACL confirmed by knee magnetic resonance imaging (MRI). Only copers were included in the current study, i.e., those with an ACLD knee for at

least three months and less than five years post-injury without functional impairment and instability during level walking. A participant candidate would be excluded if he/she had any neuromusculoskeletal disorders affecting walking, except for ACL injury or surgery of the lower limbs and lower back.

An *a priori* power analysis was conducted based on pilot results of lower-limb joint angles and moments from four participants per group using G*Power [26]. The analysis determined that a sample size of ten subjects per group would be required for two-way repeated measures ANOVA to achieve a power of 0.8 with a medium to large effect size (Cohen's $d > 0.4$) at a significance level of 0.05. Therefore, 13 subjects per group were considered adequate for the main objectives of the current study.

Experiment Protocol

In the gait laboratory of a university hospital, subjects walked at their preferred pace along a 10-meter walkway. This included either level walking or ascending a 3-meter slope with inclines of 3°, 6°, or 9°, positioned in the middle of the walkway, enabling the identification of progressive effects of slope angles. The sequence of level walking and slope conditions was determined using a random number table in a counterbalanced measures design. Each slope was constructed from five separate wooden units placed over three forceplates (AMTI, USA) located at the centre of the walkway (Figure 1). The units were arranged without touching each other, and thin anti-slip mats were attached to their bottoms to prevent sliding on the floor or forceplates. The top surfaces of the units were angled to match the required slope degrees. This setup allowed for measuring ground reaction forces (GRFs) for each lower limb at a sampling rate of 1,200 Hz. It also enabled subjects to take at least two steps on the slope before and after stepping onto the forceplates, ensuring accurate and reliable data collection.

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Each subject was carefully equipped with thirty-nine infrared retro-reflective markers placed on specific anatomical landmarks to track body segment motions [27-31]. For the pelvis and lower limbs, markers were positioned bilaterally on the anterior and posterior superior iliac spines (ASIS and PSIS), greater trochanters, mid-thighs, medial and lateral femoral epicondyles, heads of the fibula, tibial tuberosities, medial and lateral malleoli, navicular tuberosities, second metatarsal heads, fifth metatarsal bases, and heels [32-35]. For the upper body, markers were attached to the acromial processes, medial and lateral humeral epicondyles, ulnar styloids, ear canals, and the C7-T1 junction [36-38].

Three-dimensional (3D) trajectories of these markers were recorded using an eight-camera motion capture system (T40, Vicon, Oxford Metrics Group, U.K.) at a sampling rate of 200 Hz. Each subject performed trials resulting in data from three complete upslope gait cycles per limb. For subsequent statistical analysis, data from both sides were averaged for the Control group. In contrast, for the ACLD group, data were analysed separately for the affected side (ACLD-A) and the sound side (ACLD-D).

Data Analysis

Using the measured kinematic and forceplate data, inverse dynamics analyses were performed to obtain the joint angles and the joint forces and moments, i.e., intersegmental resultant forces and moments at the joint centre relative to the distal segment. Each body segment was assigned an orthogonal coordinate system with the positive x-axis pointing anteriorly, the positive y-axis upward, and the positive z-axis to the right [39]. Joint angles were derived from the joint's rotation matrix (from distal to proximal) using a Cardanic rotation sequence of z-x-y [40]. The angular motions of the trunk segment were described relative to the laboratory coordinate system. Inertial properties for each body segment were obtained using an optimisation-based method [41]. To minimise the effects of soft

tissue artefacts associated with skin markers, the global optimisation technique was applied [42]. The hip joint's centre of rotation was estimated using a functional method [36], while the centres of the knee and ankle joints were defined as the midpoints of the femoral intercondylar axis and the intermalleolar axis, respectively. All calculated joint forces were normalised to body weight (BW), and joint moments were normalised to BW and leg length (LL). The tibial inclination angle was computed as the angle between the tibial y-axis and vertical, yielding sagittal and frontal tibial inclination angles. The normal and shear components of the GRF on the slope surface and the knee joint forces were each projected onto the shank x- and z-axes. The temporospatial parameters, namely gait speed, cadence, step length, and step width, were also calculated [43]. The gait events of toe-offs and heel-strikes were determined from the measured GRF, which further defined the double-limb support (DLS) and single-limb support (SLS) phases [44].

Statistical Analysis

For level walking, independent t-tests were conducted to compare demographic data and flexor/extensor moments at the hip, knee, and ankle between the ACLD group and the Control group, as well as between the affected side of the ACLD group (ACLD-A) versus Control, and the unaffected side (ACLD-U) versus Control. For upslope walking, temporospatial parameters and all calculated three-dimensional kinematic and kinetic variables were analysed using a two-way mixed-design analysis of variance (ANOVA). This ANOVA included one between-group factor (comparing ACLD-A vs. Control, ACLD-U vs. Control, and ACLD-A vs. ACLD-U) and one within-group factor (slope angle).

In all statistical analyses, data from each calculated variable were time-averaged over gait cycles for each subject. For the Control group, data from both sides were averaged, whereas for the ACLD group, data from the affected and unaffected sides were analysed separately. The normality of all calculated

variables was assessed using the Shapiro-Wilk test, and the homogeneity of variance across groups was confirmed with Levene's test. When a significant effect of slope angle was detected, *post hoc* polynomial trend analyses were performed to examine linear trends. A significance level of $\alpha = 0.05$ was set for all statistical tests. All analyses were carried out using SPSS software (version 20; SPSS Inc., Chicago, IL, USA).

Results

Level Walking

The ACLD group was confirmed to be copers based on their walking performance. They showed no significant differences from the controls in the flexor/extensor moments at the hip, knee and ankle during level walking (Table S1).

Uphill Walking

No interactions existed between the group and slope angle factors for the temporospatial parameters or calculated kinematic and kinetic variables. The ACLD group showed significantly reduced cadence compared to Control, and both groups linearly decreased their gait speed and cadence with increasing slope angle (Table 1).

The ACLD group showed significantly increased anterior forces during initial DLS and SLS, but decreased posterior forces at the end of SLS, in both knees compared to the Control group (Figure 2 and Table S2). The affected knee also showed significantly decreased posterior forces during terminal DLS (Figure 2 and Table S2). The ACLD group showed significantly greater anterior tibial angles on the affected side at heel-strike and the end of SLS than the control group (Figure 3 and Table S3). No

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significant between-group differences existed in the tibial shear components of the normal and tangential GRF (Table S3). With increasing slope angle, both groups exhibited a linear increase in anterior force during initial DLS and SLS, as well as in posterior force during terminal DLS (Figure 2 and Table S2). They also exhibited a significant linear decrease in the tibial angles (i.e., closer to the vertical) (Figure 3 and Table S3). However, the tibial shear components of the normal and tangential GRF were increased with slope (Figure 3 and Table S3).

The ACLD group showed significantly increased trunk anterior tilt over the gait cycle of the affected limb when compared to the Control (Figure 2 and Table S4), with increased ankle dorsiflexion during terminal DLS but decreased ankle plantarflexion at contralateral toe-off (Table 2 and Figure S2). With increasing slope angle, both groups linearly increased their trunk anterior tilt over the affected side's stance phase and at both sides' toe-off (Table S4). They also linearly increased the hip flexion and ankle dorsiflexion over the entire stance phase and knee flexion over SLS and initial DLS (Table 2). However, both groups linearly decreased the hip extension and knee flexion at contralateral heel-strike and terminal DLS during uphill walking (Table 2).

Compared to the Control, the ACLD group showed significantly decreased extensor moment at the affected knee at the end of SLS, but increased internal rotator moment at the affected knee at the end of SLS, external rotator moment at the affected knee during SLS and extensor moment at both hips at the end of SLS (Figure 4 and Table S5). With increasing slope, both groups linearly increased knee abductor moments during initial DLS and SLS, and internal rotator moments during SLS and terminal DLS and extensor moments over stance phase, but decreased external rotator moments at contralateral toe-off (Table S5).

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Discussion

The current study identified the kinematic and kinetic alterations at the ACL-deficient knee, the secondary biomechanical adjustments at the hips and ankles, and the associated trunk motions in individuals with ACLD during uphill walking, as well as the influence of slope angles. Based on level walking, the current participants with ACLD were considered to be copers with knee extensor moments similar to those of healthy controls. However, uphill walking revealed significant between-group kinematic and kinetic changes, supporting the hypotheses that level-walking-defined copers would exhibit altered trunk kinematics and lower-limb joint kinematics and kinetics during uphill walking. Specifically, compared to healthy controls, the ACLD group demonstrated increased trunk anterior tilt across the stance phase, an increased anterior knee force during initial DLS and SLS, an increased anterior tibial angle with a decreased knee extensor moment around contralateral heel-strike (transition from SLS to terminal DLS), and greater hip extensor moments and ankle dorsiflexion during terminal DLS. In other words, during terminal DLS, the ACLD group reduced knee extensor moments and posterior knee forces by anteriorly shifting the line of action of the GRF through increased trunk anterior tilt, ankle dorsiflexion, and anterior tibial angle, at the expense of elevated hip extensor moments. In contrast, increased trunk anterior tilt alone was sufficient to maintain the functional stability of the injured knee during initial DLS and SLS. These results suggest a compensatory trunk anterior tilt strategy that helps maintain functional stability during uphill walking.

With increasing slope angles, both groups showed predictable biomechanical adjustments: during initial DLS and SLS, increased anterior knee forces with increased trunk anterior tilt, and, during terminal DLS, increased posterior knee forces alongside increased trunk anterior tilt, decreased hip extensor moments, and increased ankle dorsiflexion. They also showed a decreased anterior tibial angle and an increased knee extensor moment around early terminal DLS. These findings indicate that

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both groups adjust their trunk orientation and lower-limb joint mechanics in response to increased biomechanical demands imposed by uphill walking. Further, compared to the healthy controls, individuals with ACLD did not alter their compensatory strategies to mitigate instability at the injured knee. These findings suggest that the ACLD group rely on enhanced, task-specific strategies as slope angles increase, reflecting functional instability that remains hidden during level walking.

The compensatory kinematic and kinetic changes in the ACLD group during different subphases of uphill walking appeared to be closely related to the anterior-posterior knee forces required to meet the mechanical demands induced by the slope. As the slope increases, the anterior knee forces increased during initial DLS and SLS, while elevated posterior knee forces were found during terminal DLS (Tables S2). As the ankle dorsiflexion increased with slope, resulting in greater shank forward rotation and a reduced tibial inclination angle (Table S3). This adjustment, combined with increased trunk anterior tilt, helps reduce the anterior-posterior components of the GRF and, consequently, the anterior-posterior knee forces, which would otherwise be substantially higher. Nevertheless, even with this compensatory mechanism, anterior-posterior forces continue to increase as slope angle rises (Table S2). In a healthy knee, posterior knee force is provided mainly by the ACL and hamstrings, while the posterior cruciate ligament (PCL) and quadriceps contribute to the anterior knee force [45-47]. In ACL-deficient knees, without appropriate hamstrings activity, the absence of ACL contribution would require compensatory adaptations from other force-bearing structures. This was evident during terminal DLS in the ACLD group, where increased posterior forces elicited the trunk anterior tilt with other kinematic and kinetic changes at the knee and other lower-limb joints to reduce functional instability. In contrast, during initial DLS and SLS, when anterior knee forces predominated and the ACL contribution was not required, the observed increase in trunk anterior tilt was sufficient to maintain functional stability at the ACL-deficient knee.

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The current study identifies the kinematic, kinetic, and trunk adaptations of individuals with ACLD during uphill walking. During level walking, these participants showed no significant differences in knee extensor moments or lower-limb mechanics compared to healthy controls, suggesting that baseline compensatory mechanisms, such as an enhanced hamstring-to-quadriceps strength ratio, were sufficient to maintain knee functional stability. In contrast, uphill walking revealed distinct, task-specific adaptations, indicating that these baseline mechanisms were insufficient under increased biomechanical demands and that the observed strategies were necessary to preserve knee functional stability on slopes. Therefore, uphill walking serves as a sensitive functional assessment for detecting subtle compensatory mechanisms, providing a clearer distinction between copers and noncopers than level walking. Rather than serving as a binary classifier for surgical intervention, regular monitoring of uphill walking biomechanics may reveal latent control deficits or compensatory strategies not evident during level walking, offering a more sensitive tool to identify early deficits, predict risks of further knee injury or functional decline, and inform individualised treatment decisions in individuals with ACLD.

This study is the first to investigate the kinematic and kinetic adaptations of the trunk and lower limb joints in individuals with anterior cruciate ligament deficiency during uphill walking. However, it focused on slopes with inclination angles below 9°, which may limit the generalizability of the findings to steeper inclines. Nonetheless, the results remain highly applicable to real-world conditions since slopes exceeding 9° are uncommon in most daily environments. On the other hand, kinematic and kinetic adaptations were quantified using intersegmental angles and resultant forces and moments. While the current findings demonstrate inter-segment compensations, the underlying neuromuscular control mechanisms, such as specific muscle activation patterns, were not directly assessed. These measures do not isolate the contributions of individual force-bearing structures, such as muscles or ligaments, and therefore do not capture the effects of muscle co-contractions, including hamstring-

quadriceps interactions. Future studies incorporating muscle activation measurements such as electromyography and/or 3D fluoroscopy [48] are warranted to elucidate further the neuromuscular strategies underlying these observed biomechanical adaptations. Moreover, the current study was conducted exclusively on young ACLD copers; further research involving older adults or individuals with more severe functional deficits is necessary to draw broader, population-specific conclusions. Future investigations using the current experimental framework could help refine rehabilitation strategies and inform surgical decision-making for diverse populations with ACLD.

Conclusions

This study demonstrates that uphill walking uncovers hidden instability in copers with anterior cruciate ligament deficiency (ACLD) that remains undetected during level walking. While ACLD copers maintained normal gait patterns on flat surfaces, they exhibited significant biomechanical adaptations when ascending slopes. Specifically, during terminal DLS, the participants reduced knee extensor moments and posterior knee forces by anteriorly shifting the line of action of the ground reaction force through increased trunk anterior tilt, ankle dorsiflexion, and anterior tibial angle, at the expense of elevated hip extensor moments. In contrast, increased trunk anterior tilt was sufficient to maintain the functional stability of the injured knee during initial DLS and SLS. These results suggest a compensatory trunk anterior tilt strategy that helps maintain functional stability and protective mechanisms to mitigate functional instability and distribute loads more effectively across the lower limb joints. The findings highlight that traditional assessments based solely on level walking may overestimate functional stability during more demanding daily activities such as uphill walking. Incorporating uphill walking into clinical evaluations is suggested for identifying compromised movement patterns that could necessitate surgical intervention.

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Table 1 Means (standard deviations) of temporospatial parameters for participants with anterior cruciate ligament deficiency (ACLD) and healthy controls (Control) during uphill walking with 3, 6, and 9 degrees of slope angles. P-values for main slope and group effects are given as no interactions were found.

Variables	Group	Slope angle (°)			Main effects
		3	6	9	
Gait speed (m/s)	ACLD	0.90 (0.09)	0.88 (0.09)	0.86 (0.12)	$P_s = 0.043$ ↓
	Control	0.90 (0.06)	0.87 (0.07)	0.87 (0.08)	$P_g = 0.969$
Cadence (1/s)	ACLD	104.9 (9.5)	102.3 (9.0)	99.6 (12.5)	$P_s = 0.003$ ↓
	Control	112.5 (7.6)	110.1 (9.3)	109.3 (10.0)	$P_g = 0.038$*
Step width (mm)	ACLD	227.9 (36.1)	231.09 (35.9)	235.8 (37.5)	$P_s = 0.398$
	Control	212.3 (36.0)	213.58 (30.3)	211.8 (33.1)	$P_g = 0.159$
Step length (mm)	Affected side	509.8 (36.8)	514.2 (37.9)	512.5 (25.3)	$P_s = 0.340$
	Sound side	516.9 (36.0)	524.3 (42.7)	518.5 (25.4)	$P_g = 0.161$
	Control	495.2 (22.4)	500.6 (20.5)	506.2 (32.9)	

P_s = *p*-value for slope angle; P_g = ACLD vs. Control; ↓ indicates a linearly decreasing trend ($P_s < 0.05$); * indicates a significant group effect ($P_g < 0.05$).

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Table 2 Means (standard deviations) of joint angles in participants with anterior cruciate ligament deficiency (ACLD) and healthy controls (Control) during uphill walking with 3, 6, and 9 degrees of slope angles. P-values for main slope and group effects are given as no interactions were found.

Joint angles	Group	Slope angle (°)			Main effects
		3	6	9	
Hip flexion (+)/extension (-)					
Contralateral toe-off	Affected side	19.9 (6.2)	24.8 (6.0)	29.4 (6.3)	P_s = 0.001
	Sound side	21.4 (8.1)	27.7 (6.6)	31.7 (7.2)	↑
	Control	19.7 (5.7)	25.3 (5.0)	29.2 (4.9)	P_g = 0.549
Contralateral heel-strike	Affected side	-7.9 (4.8)	-5.46 (5.9)	-4.5 (7.7)	P_s = 0.001
	Sound side	-8.1 (4.2)	-6.0 (5.4)	-5.0 (6.1)	↓
	Control	-8.1 (3.8)	-6.8 (3.7)	-6.0 (4.3)	P_g = 0.885
Initial double-limb support	Affected side	23.4 (6.2)	29.3 (5.7)	34.6 (6.1)	P_s = 0.001
	Sound side	24.5 (7.5)	31.7 (6.6)	36.5 (7.6)	↑
	Control	22.2 (5.8)	28.8 (5.2)	33.7 (5.1)	P_g = 0.523
Single-limb support	Affected side	5.4 (5.4)	8.5 (5.6)	11.6 (6.5)	P_s = 0.001
	Sound side	5.1 (5.9)	9.5 (5.8)	11.4 (6.2)	↑
	Control	4.2 (4.5)	7.4 (4.3)	9.7 (4.8)	P_g = 0.728
Terminal double-limb support	Affected side	-7.4 (4.0)	-4.9 (5.3)	-3.9 (6.3)	P_s = 0.001
	Sound side	-6.9 (3.9)	-5.0 (5.3)	-4.3 (5.6)	↑
	Control	-6.2 (3.7)	-5.3 (3.6)	-4.8 (4.1)	P_g = 0.999
Knee flexion (+)/extension (-)					
Contralateral toe-off	Affected side	15.75 (7.7)	21.19 (7.1)	26.36 (8.5)	P_s = 0.001
	Sound side	18.34 (8.0)	25.75 (5.7)	32.80 (6.2)	↑
	Control	15.03 (8.4)	22.21 (7.2)	29.89 (6.6)	P_g = 0.229
Contralateral heel-strike	Affected side	13.11 (4.2)	13.22 (5.0)	11.92 (5.2)	P_s = 0.001
	Sound side	13.59 (4.6)	11.50 (5.1)	10.99 (5.4)	↓
	Control	12.1 (3.7)	10.1 (4.4)	9.71 (5.2)	P_g = 0.505
Initial double-limb support	Affected side	13.1 (7.4)	19.9 (6.6)	26.8 (7.6)	P_s = 0.001
	Sound side	15.9 (8.0)	24.7 (6.1)	32.3 (6.7)	↑
	Control	13.1 (7.9)	21.2 (6.5)	30.4 (5.9)	P_g = 0.248
Single-limb support	Affected side	12.6 (5.7)	14.4 (5.5)	17.1 (5.1)	P_s = 0.001
	Sound side	12.1 (4.9)	14.8 (4.9)	17.7 (4.4)	↑
	Control	9.7 (5.6)	11.9 (5.8)	15.5 (5.8)	P_g = 0.402
Terminal DLS	Affected side	22.0 (4.9)	21.3 (5.3)	20.0 (5.3)	P_s = 0.001
	Sound side	24.6 (5.2)	21.9 (5.8)	20.5 (5.9)	↓
	Control	23.7 (4.5)	20.8 (4.5)	19.7 (5.8)	P_g = 0.802
Ankle dorsiflexion (+)/plantarflexion (-)					
Contralateral toe-off	Affected side	-1.0 (3.9)	5.0 (3.4)	11.4 (6.0)	P_s = 0.001
	Sound side	0.4 (4.1)	6.4 (3.53)	14.1 (4.2)	↑

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Contralateral heel-strike	Control	-0.9 (4.0)	5.1 (3.1)	15.2 (9.0)	P_g = 0.443
	Affected side	11.9 (4.2)	14.0 (5.0)	18.1 (6.5)	P_s = 0.001 ↑
	Sound side	12.0 (4.0)	14.2 (4.1)	15.4 (5.6)	
	Control	8.3 (3.7)	10.2 (3.5)	13.5 (9.4)	P_g = 0.091
Toe-off	Affected side	-8.1 (4.6)	-7.2 (7.6)	-5.2 (9.5)	P_s = 0.843 P_g = 0.013*
	Sound side	-9.6 (5.5)	-8.7 (5.0)	-10.7 (7.2)	
	Control	-12.4 (4.0)	-12.9 (3.2)	-14.7 (6.9)	
Initial double-limb support	Affected side	-3.2 (3.7)	1.5 (3.9)	7.0 (4.8)	P_s = 0.001 ↑
	Sound side	-1.9 (4.3)	4.1 (3.7)	10.4 (4.0)	
	Control	-3.2 (3.8)	2.2 (2.9)	11.9 (8.9)	P_g = 0.271
Single-limb support	Affected side	7.8 (3.5)	11.2 (2.9)	15.2 (3.4)	P_s = 0.001 ↑
	Sound side	7.9 (3.5)	11.9 (3.6)	16.2 (3.7)	
	Control	6.2 (3.6)	9.9 (3.1)	16.6 (9.2)	P_g = 0.768
Terminal double-limb support	Affected side	4.2 (4.9)	4.9 (5.9)	7.1 (6.4)	P_s = 0.007 ↑
	Sound side	4.3 (4.4)	6.1 (4.4)	6.3 (5.9)	
	Control	0.5 (3.3)	1.6 (3.3)	2.9 (6.3)	P_g = 0.039*

P_s = *p*-value for slope angle; P_g = ACLD vs. Control; ↑ indicates a linearly increasing trend and ↓ indicates a linearly decreasing trend (P_s < 0.05); * indicates a significant difference between the affected side of ACLD and Control group.

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Figure Captions

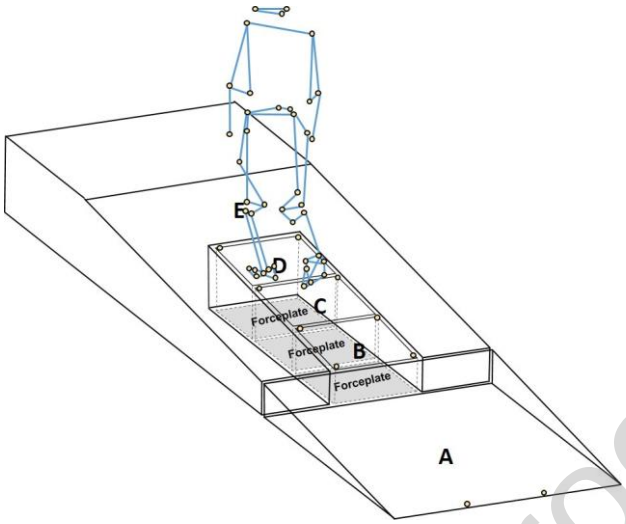

Figure 1 Schematic diagram of the sloped walkway, forceplates, and a simplified figure of a participant with the marker set, produced based on a photograph of a typical trial. The walkway consisted of four separate wooden units (A-E). The first unit (A), a wedge, was positioned in front of the first forceplate, followed by three wooden blocks (B-D), each placed on a single forceplate. The first unit, a wedge (A), was positioned in front of the first forceplate, while three wooden blocks (B, C, and D) were placed on the two forceplates. The final unit (E) consisted of a U-shaped wedge that encircled the blocks B-D from behind block A, creating a continuous slope with a horizontal surface at the end. The top surfaces of the blocks were aligned to maintain a consistent slope, and each block was positioned to ensure there was no contact with the others.

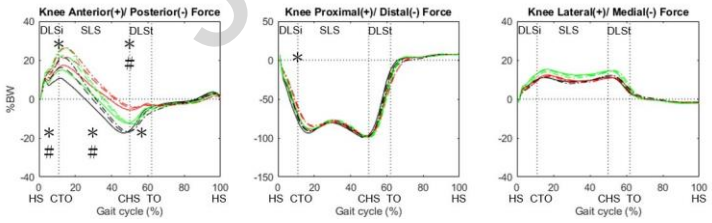

Figure 2 Ensemble-averaged three-dimensional knee joint forces at the lower limbs of the affected side (red lines) and the sound side (green lines) of the ACLD and Control groups (black lines) during 3° (solid lines), 6° (dashed lines), and 9° (dashed-dotted lines) uphill walking (HS: heel-strike; CTO: contralateral toe-off; CHS: contralateral heel-strike; TO: toe-off; SLS: single-limb support; DLSi :

initial double-limb support; DLSt: terminal double-limb support; * : affected side of the ACLD significantly different from Control; # : sound side of the ACLD significantly different from Control). P values are provided in Table S2.

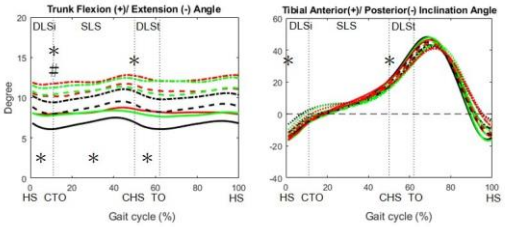

Figure 3 Ensemble-averaged trunk kinematics and tibial inclination angles in the sagittal plane of the affected side (red) and the sound side (green) of the ACLD and Control groups (black lines) during 3 ° (solid lines), 6° (dashed lines), and 9° (dashed-dotted lines) uphill walking (HS: heel-strike; CTO: contralateral toe-off; CHS: contralateral heel-strike; TO: toe-off; SLS: single-limb support; DLSi : initial double-limb support; DLSt: terminal double-limb support; * : affected side of the ACLD significantly different from Control; # : sound side of the ACLD significantly different from Control). P values are provided in Tables S3 and S4.

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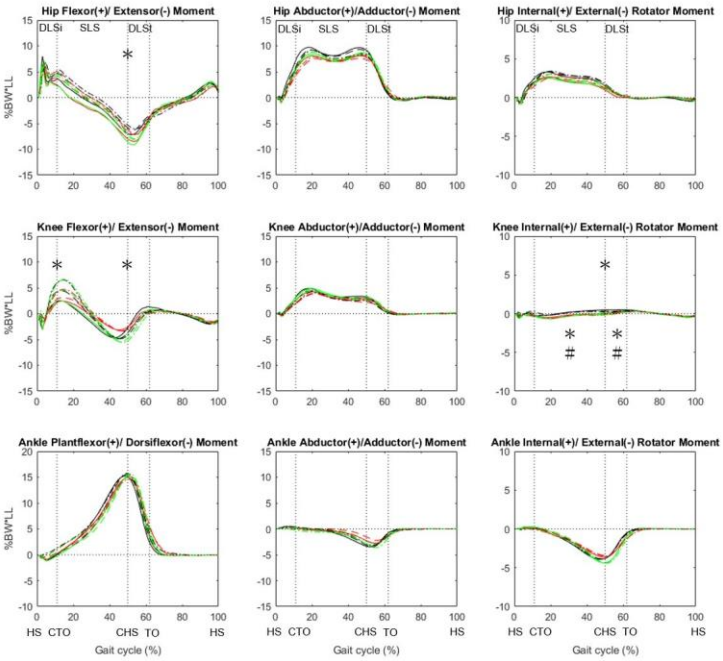

Figure 4 Ensemble-averaged three-dimensional joint moments at the lower limbs of the affected side (red lines) and the sound side (green lines) of the ACLD and Control groups (black lines) during 3° (solid lines), 6° (dashed lines), and 9° (dashed-dotted lines) uphill walking (HS: heel-strike; CTO: contralateral toe-off; CHS: contralateral heel-strike; TO: toe-off; SLS: single-limb support; DLSi : initial double-limb support; DLSt: terminal double-limb support; * : affected side of the ACLD significantly different from Control; # : sound side of the ACLD significantly different from Control). P values are provided in Table S5.

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Declarations

Ethics Approval and Consent to Participate

Ethical approval was obtained from the TMU-Joint Institutional Review Board (Approval Number: N202004053). All study participants gave verbal and written consent. The study was conducted following the Declaration of Helsinki.

Competing Interests

The authors have no competing interests to declare.

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Authors' Contributions

J-YW: data curation, resources and writing — original draft.

S-HL: conceptualisation, data curation, resources, formal analysis, and writing— original draft.

C-HH: resources and formal analysis.

K-WW: data curation and writing — original draft.

T-MW: conceptualisation and writing — review and editing.

T-WL: conceptualisation, supervision, writing — original draft, and writing — review and editing.

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Highlights

- Uphill walking reveals hidden functional instability in ACLD copers, undetected on level walking

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- ACLD copers exhibited trunk anterior tilt to reduce knee loads to mitigate instability
- Uphill walking aids clinical evaluation and informs surgical decisions in ACLD copers

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